



North South University

Department of Electrical & Computer Engineering

Project Report

ROAD HERO

Smart Road Hazard Detection & Community Safety System

Section: 04

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Project Overview

Road Hero is a community-driven mobile application designed to improve road safety across urban and semi-urban environments. The application empowers ordinary citizens to actively participate in identifying, reporting, and monitoring road hazards in real-time. By combining advanced artificial intelligence with crowdsourced community action, Road Hero creates a comprehensive, self-sustaining safety network that connects drivers, pedestrians, and road authorities. The core philosophy of Road Hero is that safer roads are built not just by governments, but by informed and engaged communities. Every user becomes a "Road Hero" contributing to a living map of road conditions that benefits all travelers in their area.

Problem Statement

Road infrastructure in Bangladesh and many developing nations faces serious deficiencies that are inadequately tracked or communicated. The following core problems motivated the development of Road Hero:

- Poor road conditions cause thousands of accidents annually, putting drivers and pedestrians at serious risk of injury and property damage.
- No real-time hazard reporting system exists for drivers to warn others about sudden road dangers such as potholes, debris, or flooding.
- Road damage tracking remains fragmented with no centralized platform connecting citizens, authorities, and maintenance teams.
- Citizens have no structured way to hold authorities accountable for unrepaired road damage.

Technologies Used

Frontend — Flutter

The Road Hero mobile application is built using Flutter, Google's open-source UI toolkit for building natively compiled applications from a single codebase. Flutter enables cross-platform deployment on both Android and iOS, ensuring the app reaches the widest possible user base.

- Language: Dart
- Framework: Flutter (cross-platform mobile)
- UI: Material Design components with custom theming (Fresh Emerald, Obsidian Dark, Arctic Frost, etc.)
- Maps: Interactive map integration with real-time hazard pin rendering
- Camera: Native camera access for photo capture and AI scan submission
- State Management: Reactive state handling for live updates

Backend & Database — Firebase

The entire backend infrastructure is powered by Google Firebase, providing a scalable, real-time, serverless backend solution. Firebase's suite of tools handles authentication, data storage, cloud functions, and push notifications.

Firestore Service	Purpose in Road Hero
Firestore Authentication	Secure user login/signup with email-password and Google OAuth. Separate authentication flows for Citizens and Admins.
Cloud Firestore	Real-time NoSQL database storing hazard reports, user profiles, community posts, verification votes, leaderboard data, and chat messages.
Firestore Storage	Cloud storage for hazard evidence photos uploaded by users during hazard reporting.
Firestore Cloud Messaging	Push notifications for hazard proximity alerts, community messages, and official hazard broadcasts.
Firestore Cloud Functions	Serverless backend logic for triggering hazard resolution (auto-removal after 3 votes), point calculations, and admin workflows.

AI Integration — Google Gemini API

Road Hero integrates Google's Gemini AI (specifically Gemini 1.5 Flash) for intelligent road hazard analysis. When a user submits a photo, the AI performs multimodal analysis combining the image, GPS metadata, and contextual prompts to automatically identify the type of road damage and its severity level.

- Model Used: gemini-1.5-flash (primary), with fallback to gemini-2.5-flash and gemini-pro-vision
- Analysis: Detects potholes, cracks, alligator fatigue cracking, flooding, and debris
- Output: Damage type classification, severity rating (Low / Medium / High / Severe), and descriptive analysis in English and Bangla
- False Positive Filtering: AI validates image content before posting to the community feed

Application Features

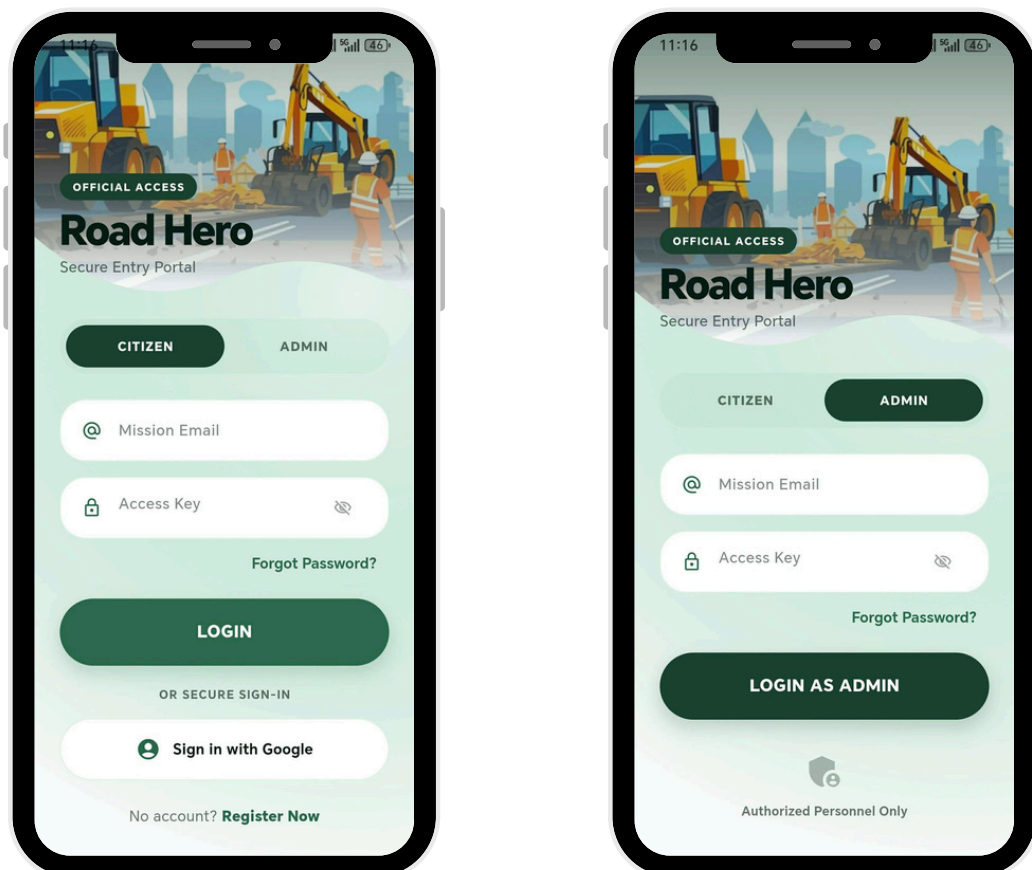
Road Hero provides a rich set of features organized around a central dashboard. All features are accessible from the homepage and are designed to work in concert to create a complete road safety ecosystem.

User & Admin Login

The authentication system is the entry point for all users. Road Hero provides two separate login portals to distinguish between regular citizens and system administrators.

Working Mechanism:

- Citizens login using their registered email and password, or via Google OAuth single sign-on.
- Admins use a separate secure portal restricted to authorized personnel only, identified by a dedicated "Login As Admin" button.
- Both portals are secured using Firebase Authentication with modern encryption standards.
- Multi-factor authentication options are available to ensure account safety while maintaining a seamless login experience.
- Once authenticated, users are routed to role-appropriate dashboards: Citizens to the homepage feature menu, Admins to the Command Center.

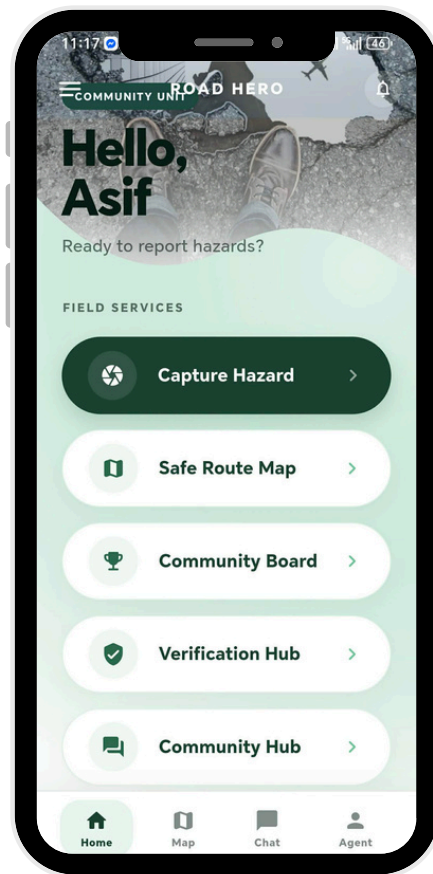


Homepage Central Command Dashboard

The homepage serves as the main navigation hub for all core features in the application.

Working Mechanism:

- Upon login, users see a personalized greeting (e.g., 'Hello, Asif Ready to report hazards?').
- Five field service tiles are displayed: Capture Hazard, Safe Route Map, Community Board, Verification Hub, and Community Hub.
- Each tile provides one-tap navigation to the corresponding feature.
- A real-time notification system keeps users informed of nearby hazards as they are reported by other community members.
- The bottom navigation bar provides persistent access to Home, Map, Chat, and Agent (Profile) sections.

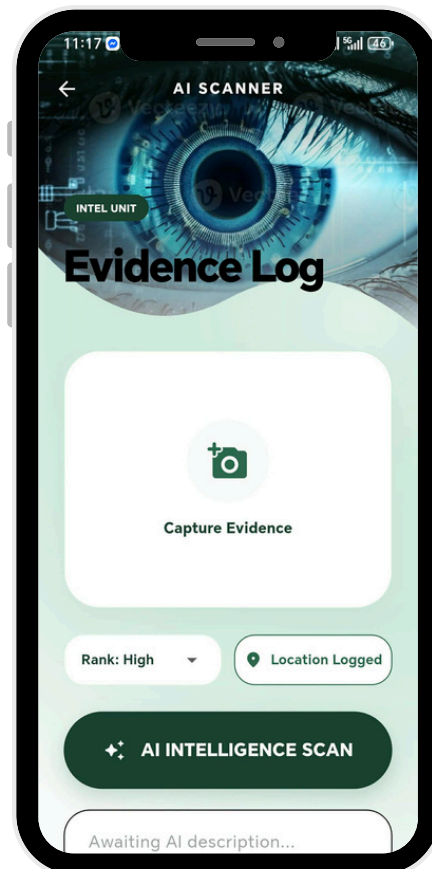


Capture Hazard AI-Powered Reporting

This is the primary reporting feature of RoadHero, enabling users to document and submit road hazards using their smartphone camera.

Working Mechanism:

- The user opens the AI Scanner screen and taps 'Capture Evidence' to take a photo or upload an existing image of the road damage.
- The app automatically captures GPS coordinates and timestamps the submission.
- On pressing 'AI Intelligence Scan', the image is encoded in Base64 and sent to the Google Gemini 1.5 Flash API along with a structured prompt and location metadata.
- The AI performs multimodal analysis and returns a hazard description including damage type (e.g., Severe Potholes, Alligator Cracking) and severity rating (Low / Medium / High / Severe).
- The report is automatically posted to the Community Hub feed, pinned on the Safe Route Map, and added to the Verification Hub queue for community validation.
- The submitting user earns points, contributing to their Community Board rank.

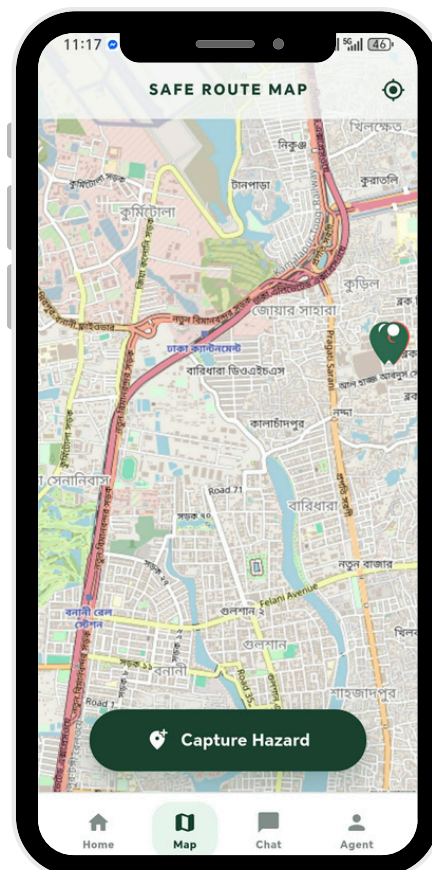


Safe Route Map — Live Hazard Navigation

The SafeRoute Map provides a real-time interactive map of all active road hazards reported by the community.

Working Mechanism:

- The map loads using GPS-integrated mapping and displays hazard pins at their exact GPS-logged locations.
- Color-coded markers indicate hazard severity; users can quickly identify low, medium, high, and critical hazards.
- The map updates in real-time as new hazards are reported and as resolved hazards are removed from the system.
- Users can plan their routes with hazard awareness, avoiding dangerous areas and accident-prone zones.
- Only community-verified hazard locations appear with full detail; unverified reports appear as provisional markers.
- A 'Capture Hazard' shortcut button is accessible directly from the map screen.

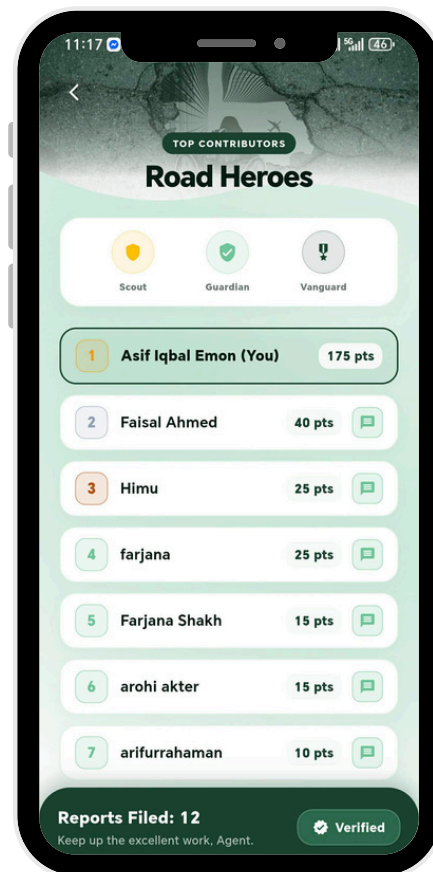


Community Board — Gamified Leaderboard

The CommunityBoard gamifiesroadsafety participationbyrewarding users with points and ranks for their contributions.

Working Mechanism:

- Users earn Experience Points (XP) for every hazard they report, every verification vote they cast, and every hazard they help resolve.
- Points accumulate to determine a user's Rank starting from Scout, progressing to Guardian and Vanguard.
- The leaderboard displays all community members sorted by XP, with the current user's position highlighted.
- Top contributors are publicly recognized, motivating active and sustained participation.
- A running count of total reports filed and verified status is shown at the bottom of each user's leaderboard card.

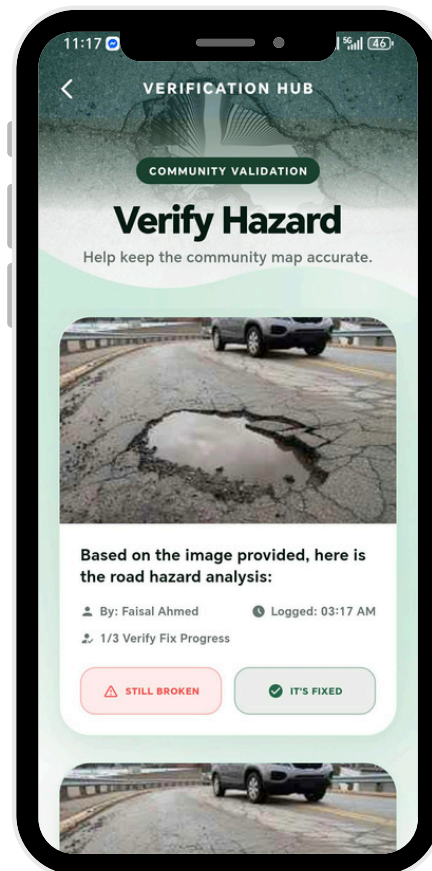


Verification Hub — Crowdsourced Validation

The VerificationHub is the quality control layer of the Road Hero system, ensuring all hazard data on the map is accurate and up-to-date.

Working Mechanism:

- All hazard reports submitted to the system are queued in the Verification Hub for community review.
- Community members who encounter the reported road location can cast a vote: 'Still Broken' (hazard persists) or 'It's Fixed' (hazard resolved).
- A progress indicator shows how many of the 3 required resolved votes have been received (e.g., '1/3 Verify Fix Progress').
- When 3 resolved votes are collected from verified users, a Firebase Cloud Function automatically updates the hazard status to 'Resolved' and removes it from the active map and feed.
- This crowdsourced validation model ensures only current, real hazards remain visible to drivers.
- Users earn XP for each verification vote they cast, regardless of the outcome.

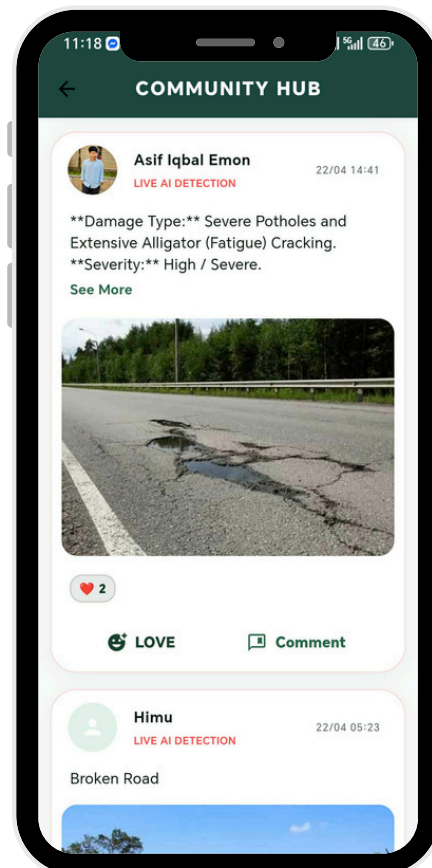


Community Hub — Social Safety Feed

TheCommunityHubtransforms roadsafetyreportinginto a social experience, enabling users to engage with each other's hazard submissions.

Working Mechanism:

- Every hazard reportedthrough Capture Hazard is automatically published as a post in the CommunityHubfeed.
- Posts display thereporter's name, timestamp, AI-generated hazard description, and the submitted photo.
- Community memberscan react to posts with a 'Love' reaction and leave comments to provide additionalcontext or updates.
- Engagement withpostsincreases their visibility, helping the community prioritize the most critical andfrequently validated hazards.
- Posts tagged with'LiveAI Detection' indicate hazards that have been analyzed by the Gemini AI system.

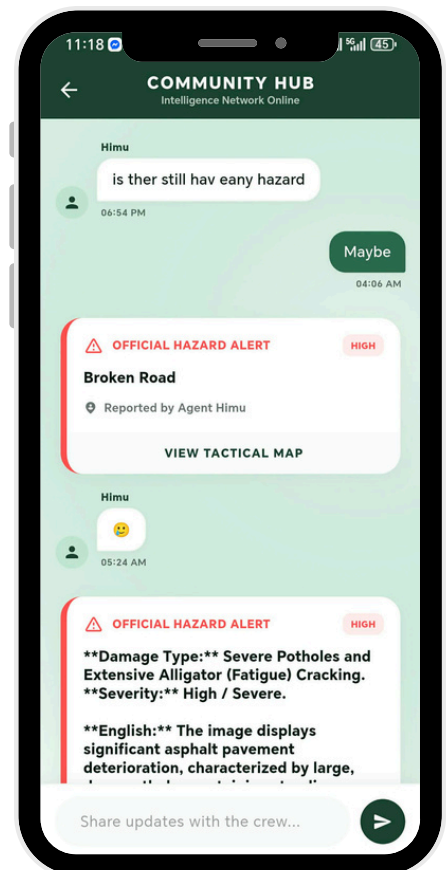
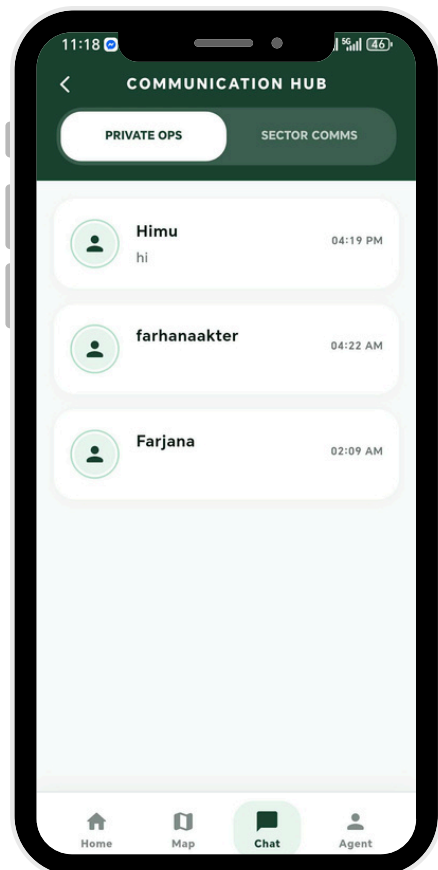


Chat System — Real-Time Communication

The Chat System provides direct and group communication channels for community members to coordinate around road safety issues.

Working Mechanism:

- **Private Ops:** One-to-one private messaging between any two registered users for direct communication about specific hazards or safety concerns.
- **Sector Comms:** A community group chat where all users in the network can participate in neighborhood-wide discussions and receive real-time hazard alerts.
- **Official Hazard Alert cards** are automatically injected into the group chat when high-severity hazards are reported, including the damage type, severity level, and a 'View Tactical Map' link.
- **Firebase Cloud Messaging** powers instant notification delivery so users are alerted even when the app is in the background.

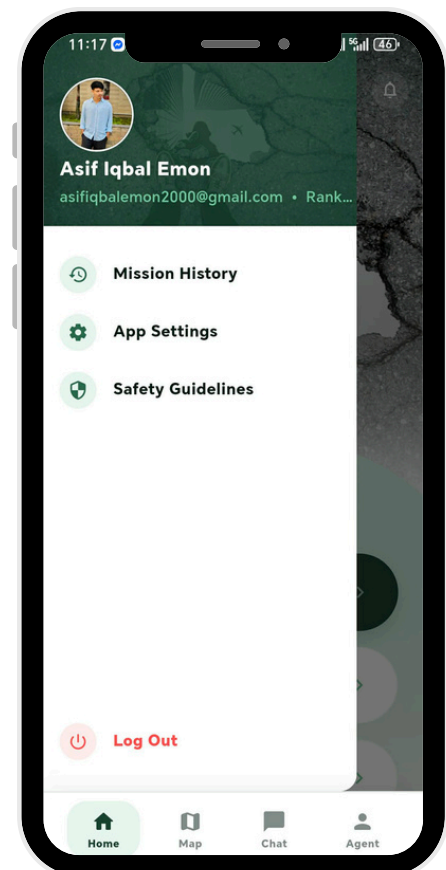
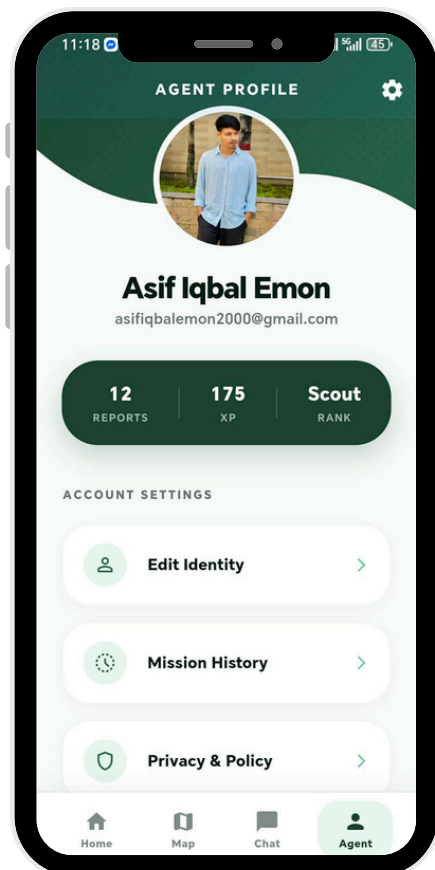


Agent Profile — Personal Dashboard

The Agent Profile serves as the user's personal account management center within Road Hero.

Working Mechanism:

- Displays the user's profile photo, full name, email address, total reports filed, accumulated XP, and current rank badge.
- Edit Identity: Allows users to update their display name, profile photo, and other personal details.
- Mission History: Provides a complete log of all hazards the user has ever reported or verified, with status indicators.
- Privacy & Policy: Links to the application's privacy policy and usage terms.
- App Settings: Access to notification preferences, theme customization, and device interaction settings.
- Safety Guidelines: Reference documentation for how to correctly identify and report different types of road hazards.

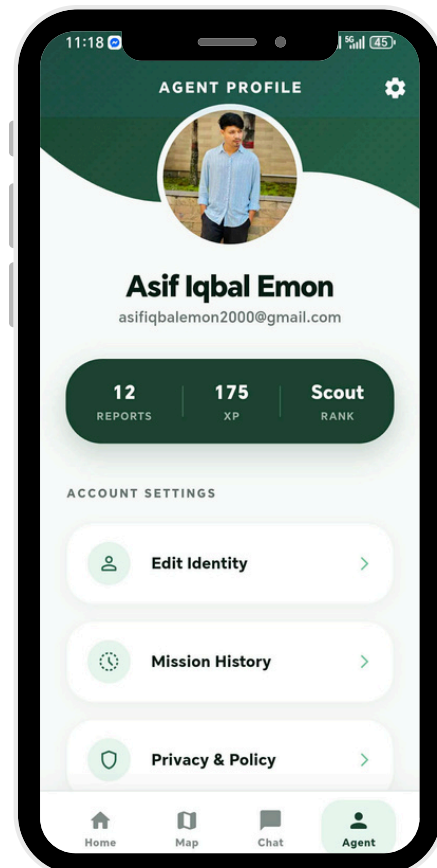
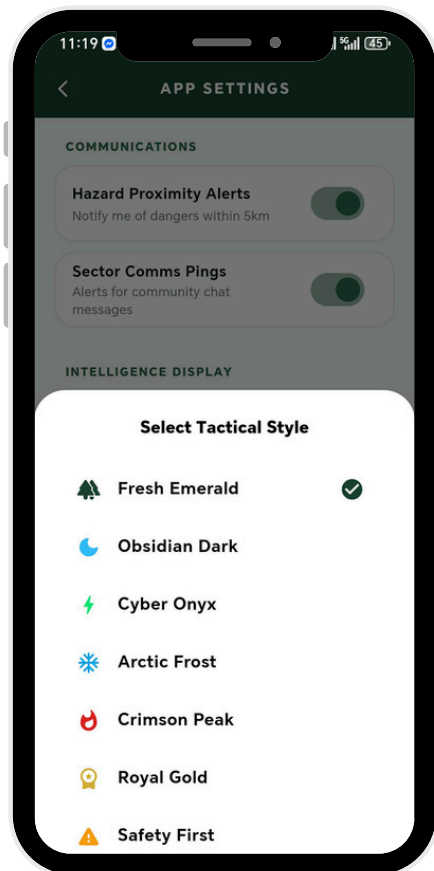


Settings Menu — Customization & Configuration

The Settings Menu gives users control over their app experience and device integration preferences.

Working Mechanism:

- **Hazard Proximity Alerts:** Toggle to enable push notifications when a new hazard is reported within 5km of the user's current location.
- **Sector Comms Pings:** Toggle to receive alerts for new community chat messages.
- **Tactical Style (Theme):** Select from multiple color themes: Fresh Emerald, Obsidian Dark, Cyber Onyx, Arctic Frost, Crimson Peak, Royal Gold, Safety First to personalize the app's visual appearance.
- **Map Configuration:** Configure which map tile style is used in the Safe Route Map screen.
- **Restore Hidden Hazards:** Bring previously archived hazard reports back to the active map display.
- **Clear Asset Cache:** Free up device storage by removing cached hazard images.

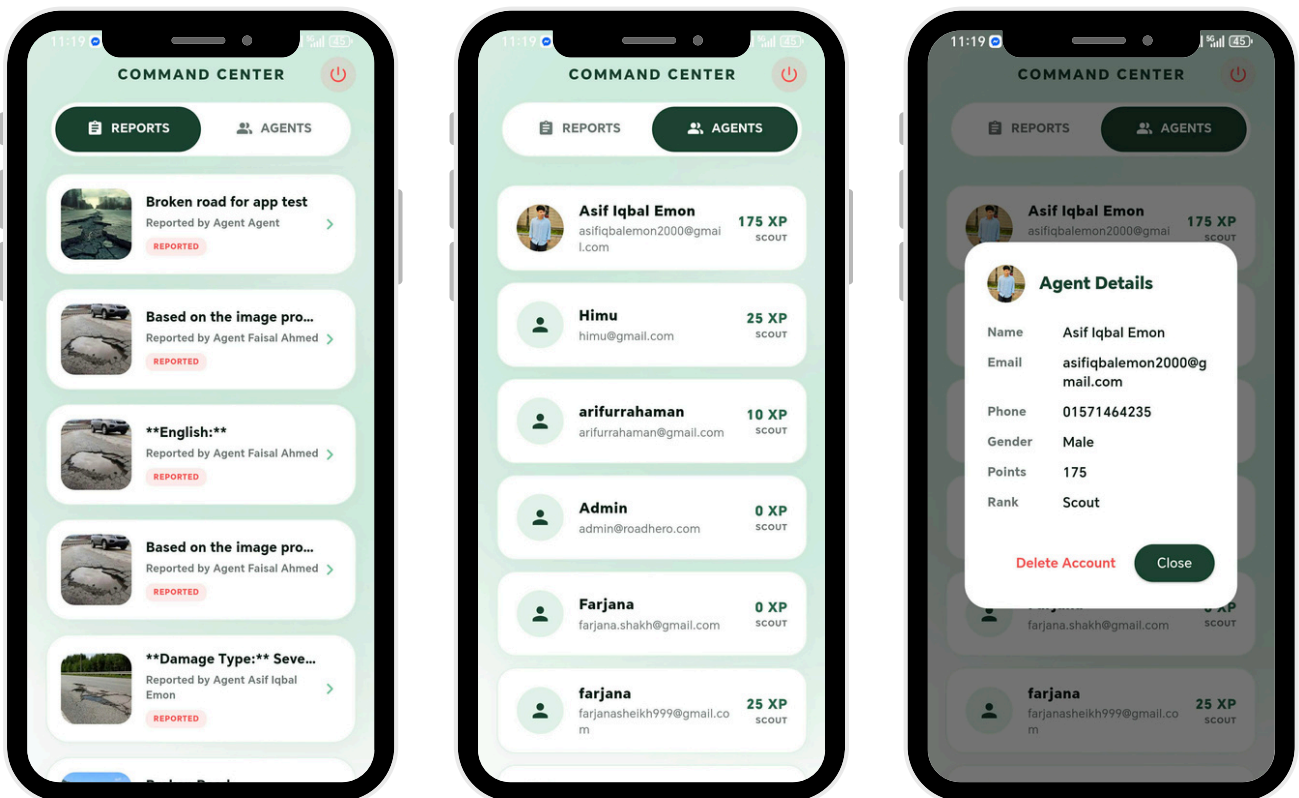


Admin Panel — Command Center

The Admin Panel provides system administrators with a comprehensive dashboard to oversee all platform activity and manage hazard reports and user accounts.

Working Mechanism:

- **Reports Tab:** Displays all hazard reports submitted platform-wide, with thumbnails, reporter names, AI descriptions, dates, severity levels, and current status badges (Reported/InProgress/Resolved).
- **Admins can tap any report to expand full details** including the complete AI analysis in both English and Bangla and update the hazard workflow status using three action buttons: Reported, Progress, Resolved.
- **Agents Tab:** Lists all registered users with their email, XP, and rank. Admins can tap any user to view their complete profile including phone number, gender, and point total, with an option to delete the account if necessary.
- All admin actions are logged and reflected in real-time across the entire platform through Firestore's live sync.



Future Roadmap

RoadHero is designed as an evolving smart city platform. The following enhancements are planned for future development cycles:

- **Government Integration:** Direct API connection to municipal road maintenance systems, enabling automatic work order generation when hazards reach a defined severity threshold. This closes the loop between citizen reporting and official repair action.
- **Live Traffic Integration:** Combining hazard data with real-time traffic flow information to provide dynamic route optimization and congestion alerts, similar to Waze but focused specifically on road damage.
- **Advanced AI Accuracy:** Training custom road hazard detection models on locally collected Bangladeshi road condition datasets to improve classification accuracy beyond general-purpose vision models.
- **Smart Notification Routing:** Proactive hazard alerts delivered based on the user's saved routes and regular travel patterns, notifying them of new hazards along their commute before they set out.
- **Web-Based Admin Dashboard:** A full browser-based administrative interface with analytics charts, heatmaps, and export capabilities for government and municipal use.

Conclusion

RoadHero demonstrates how modern mobile technology, artificial intelligence, and community engagement can be combined to address a critical public safety challenge. By putting the power of road hazard detection in the hands of every citizen, and reinforcing it with AI analysis and a structured verification system, Road Hero creates a self-improving safety network that benefits all road users.

The application successfully integrates Flutter for cross-platform mobile delivery, Firebase for scalable real-time backend infrastructure, and Google Gemini AI for intelligent image analysis — resulting in a cohesive, production-ready system. As a scalable smart city solution, Road Hero is positioned for integration with government road maintenance authorities and urban infrastructure systems, with the ultimate vision of making roads safer for everyone.

— End of Report —

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